

Cyber Security & Forensics II (05201352)

Q1 (a) Short Questions (1 Mark Each)

(i) Define CIA triad in cybersecurity.

👉 CIA triad stands for **Confidentiality, Integrity, and Availability** – the three core principles of cybersecurity ensuring data protection, accuracy, and accessibility.

(ii) What is the purpose of virtualization?

👉 Virtualization allows multiple operating systems or applications to run on a single physical machine by creating **virtual environments**, improving resource utilization and security.

(iii) Give one example of symmetric encryption.

👉 **AES (Advanced Encryption Standard)** is a widely used symmetric encryption algorithm.

Q1 (b) MCQs / True-False / Fill in Blanks (1 Mark Each)

1. Which tool is commonly used for port scanning in Kali Linux?

✓ a) Nmap

2. True or False: Steganography hides information in encrypted files. Justify.

👉 **False** – Steganography hides data inside other media (like images/audio), not encrypted files.

3. Which port number is used by SFTP?

✓ b) 22

4. Fill in the blank: Encryption is used to convert plaintext into **ciphertext**.

5. Which of these is a feature of a firewall?

✓ b) Filter traffic

6. What does IDS stand for in cybersecurity?

✓ c) Intrusion Detection System

7. What is the use of hashing?

👉 To **store passwords securely** and ensure data integrity.

Q2 (a) Two Questions of 2 Marks

(i) What is a vulnerability? Mention one example.

👉 A **vulnerability** is a weakness in a system, network, or application that attackers can exploit.

- ♦ Example: **SQL Injection vulnerability** in a web application.

(ii) Describe any two types of cyber threat actors.

1. **Hacktivists** – Attackers motivated by political or social agendas.
 2. **Insiders** – Employees or contractors who misuse access to harm the organization.
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Q2 (b) Two Questions of 3 Marks

(i) Explain how firewalls help prevent unauthorized access.

- Firewalls act as a **barrier between internal and external networks**.
- They filter traffic based on rules (IP, port, protocol).
- Prevent malicious packets, block suspicious IPs, and enforce access control.

(ii) Describe the role of Burp Suite in web application testing.

- Burp Suite is a **penetration testing tool** for web security.
 - Used for **scanning vulnerabilities**, intercepting requests, testing SQL injection, XSS, etc.
 - Helps security testers identify and exploit weaknesses.
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Q3 Attempt Any TWO (5 Marks Each)

(i) Different types of firewalls with examples:

1. **Packet Filtering Firewall** – Filters based on IP/port (e.g., Cisco ACLs).
2. **Stateful Inspection Firewall** – Tracks session state (e.g., Check Point).
3. **Proxy Firewall** – Acts as intermediary, hides internal systems (e.g., Squid Proxy).
4. **Next-Gen Firewall (NGFW)** – Includes intrusion prevention, deep packet inspection (e.g., Palo Alto).

(ii) Working of digital signatures in ensuring integrity:

- A digital signature is created using **hash + private key encryption**.
- The receiver verifies it with the sender's **public key**.

- Ensures **data integrity, authenticity, and non-repudiation**.

(iii) Importance of secure coding practices:

- Prevents vulnerabilities like SQL injection, XSS, buffer overflow.
 - Improves system reliability and compliance.
 - Example: Using prepared statements instead of string concatenation in SQL queries.
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Q4 (a) Case Study (5 Marks)

Scenario: Large files transferred are lost/corrupted due to TTL and Fragment Offset issues.

- **TTL (Time To Live):** Prevents packets from looping indefinitely by reducing counter at each hop.
- **Fragment Offset:** Indicates packet position when large data is split into fragments.

If misconfigured:

- Packets may expire before reaching destination.
- Fragment reassembly may fail → data corruption/loss.

Impact: Network communication becomes unreliable.

Solution: Proper configuration of TTL values, avoiding fragmentation via **Path MTU**

Discovery, using secure routing.

Q4 (b) (5 Marks)

Option 1: TCP vs UDP in different services

- **File Transfer (FTP, HTTP) → TCP** (reliable, ordered delivery).
- **Video Streaming (YouTube, VoIP) → UDP** (fast, tolerates packet loss).
- **DNS Resolution → UDP** (fast lookups), but sometimes **TCP** for large queries.

✓ Recommendation:

- File transfer → TCP
 - Streaming → UDP
 - DNS → UDP (with fallback to TCP)
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Option 2: Static vs Dynamic Routing

- **Static Routing:** Manual, simple, secure, good for small networks, but not scalable.
- **Dynamic Routing:** Uses protocols like OSPF, RIP, BGP; adapts to changes automatically, scalable, but consumes resources.

✓ For a **medium-sized, growing network**, **Dynamic Routing** is better due to scalability and automatic updates.